Essays on Heterogeneous Returns, Trust, and Household Wealth

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Roadmap

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Introduction

1. **Chapter 1**: Can return heterogeneity help a standard heterogeneous-agent model match empirical wealth moments?
2. **Chapter 2**: Is there a systematic relationship between trust and returns to net wealth, and can we still estimate a persistent household return component?
3. **Chapter 3**: If wealth is tied to social institutions and historical group structure, does the standard objective benchmark understate wealth inequality?

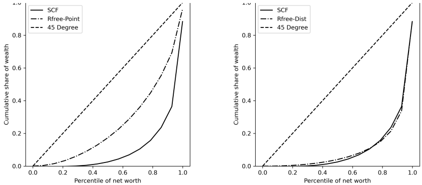
Taken together, the chapters move from explaining wealth concentration, to explaining realized performance, to measuring inequality itself.

Chapter 1: Matching Wealth Moments with Heterogeneous Returns

- I incorporate return heterogeneity into a standard heterogeneous-agent model and estimate the return distribution needed to match the wealth distribution.
- Labor income uncertainty and life-cycle structure explain part of wealth dispersion, but not enough to reproduce the empirical skewness of wealth.
- Adding return heterogeneity yields a much better fit to SCF wealth moments and implies a more realistic aggregate MPC.

Empirically, this is motivated by evidence that realized household returns vary widely across people (Fagereng et al. 2020; Daminato and Pistaferri 2024).

Chapter 1: Main Result



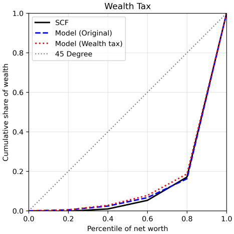
Heterogeneous returns produce a Lorenz curve much closer to the SCF target.

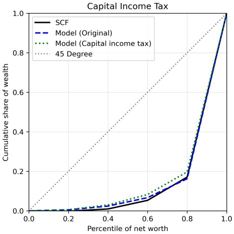
- Estimated uniform return distribution:

$$R = 1.0204, \qquad \nabla = 0.06833.$$

- Top 5% wealth share:
  - Data: 57.4%
  - No heterogeneity: 23.3%
  - Heterogeneity: 58.8%

Chapter 1: Policy Implications — Wealth Distribution





- Two revenue-equivalent policies (each raising 1% of GDP): wealth tax $1 + r - \tau_w$ lowers the gross return for all; capital income tax $1 + (1 - \tau_{ci})r$ taxes only positive returns.
- In partial equilibrium, the CIT reduces wealth concentration modestly more than the WT in both the PY and LC calibrations.
- Intuition: low-return households face near-zero CIT (little positive capital income to tax) but a positive WT regardless of their return.

Chapter 1: Policy Implications — Welfare

- Welfare is measured by consumption-equivalent Δ: the % increase in consumption under the WT that makes a newborn — before drawing their return type — indifferent to the CIT. $\Delta > 0$ means CIT preferred.
- **Aggregate:** $\Delta = 0.20\%$ (PY) and 0.15% (LC) — the CIT is preferred on average for a newborn.
- **By return type:** 6 of 7 types prefer the CIT; only the highest-return type ($R = 1.079$) prefers the WT. Low-return types gain most from switching to CIT because the WT reduces their already-low gross return while generating little revenue from them.

	Type 1	**Type 2**	**Type 3**	**Type 4**	**Type 5**	**Type 6**	**Type 7**
Baseline R	0.962	0.981	1.001	1.020	1.040	1.059	1.079
CE Δ (WT vs CIT)	+0.23%	+0.27%	+0.34%	+0.32%	+0.30%	+0.24%	−0.31%

Chapter 2: Moderate Trust and Maximum Performance on Financial Investments

- Economic performance is hump-shaped in trust: moderate trust maximizes outcomes at the individual level.
- Using the RAND HRS panel — which includes both household financial data and a generalized trust measure — we test whether this holds for returns to net wealth.
- It does: the return-maximizing trust level is 6–7 on a 1–10 scale, and the panel structure allows us to estimate a persistent household return component.

Chapter 2: Main Result

- Trust and Trust^2 are jointly significant: returns to net wealth are hump-shaped in trust.
- The result is robust across pooled OLS and correlated random effects (CRE/Mundlak) specifications.

	Pooled OLS		CRE	
	(1) Baseline	(2) Extended	(3) Baseline	(4) Extended
Trust	0.03***	0.03***	0.03***	0.03***
	(0.01)	(0.01)	(0.01)	(0.01)
Trust^2	−0.00**	−0.00**	−0.00***	−0.00***
	(0.00)	(0.00)	(0.00)	(0.00)
N	2,919	2,899	2,919	2,899
R^2	0.04	0.04	0.15	0.15
ρ			0.16	0.15
σ_u			0.16	0.15
σ_e			0.36	0.36

Controls: age, wealth, education, year; col. 2 & 4 add demographics.

Where's the max?

Chapter 2: Quadratic Trust Term — Where Does the Max Come From?

Starting from the panel regression with controls held fixed,

$$r_{it}^{net} = \alpha_i + \lambda_t + X_{it}'\beta + \beta_1\ Trust_i + \beta_2\ Trust_i^2 + \varepsilon_{it},$$

the implied marginal effect of trust on returns is

$$\frac{\partial r_{it}^{net}}{\partial Trust_i} = \beta_1 + 2\beta_2\ Trust_i.$$

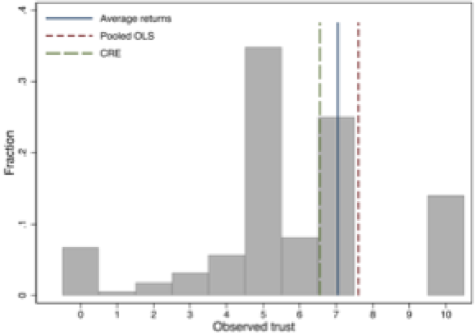
Setting this derivative to zero gives the critical point

$$Trust^* = -\frac{\beta_1}{2\beta_2}.$$

When $\hat{\beta}_1 \geq 0$ and $\hat{\beta}_2 \leq 0$, the fitted relationship in trust is concave (hump-shaped), so $Trust^*$ is a maximum: returns rise in trust up to $Trust^*$ and fall beyond it. Plugging in the estimated coefficients gives the values reported on the previous slide (6–7 across specifications).

Back to Main Result

Chapter 2: Return-Maximizing Trust Level



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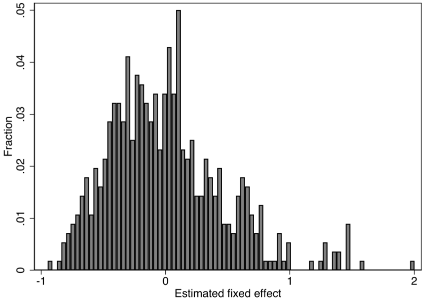
Specification	No controls	With controls
Avg. returns	6.70	7.07
Pooled OLS	6.89	7.61
CRE	6.34	6.59

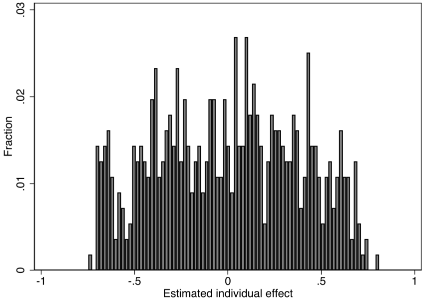
- Estimated return-maximizing trust is consistently 6–7 across all specifications.
- The observed trust distribution concentrates near this optimal range.

Chapter 2: Persistent Return Component

$\rho \approx 0.16$, $\sigma_u \approx 0.16$ (persistent SD), $\sigma_e \approx 0.36$ (idiosyncratic SD): roughly 16% of total return variance reflects stable household differences, unchanged by adding portfolio-share controls.



FE: estimated individual effects (full controls).



CRE: estimated individual effects (full controls).

Chapter 3: Wealth Inequality Under Ambiguity

- Motivated by Rawls' discussion of justice behind a veil of ignorance, I propose a wealth inequality measure based on choice under ambiguity.
- The key idea is that wealth distributions may need to be ranked using socially relevant states, rather than objective uncertainty alone.
- I then adapt the social-welfare approach to inequality measurement to define an ambiguity-based wealth inequality measure.

Question: does the standard objective benchmark understate wealth inequality? Can we reasonably propose a measure of inequality which is specific to the historical context of wealth accumulation in the U.S.? If so, does it more closely track other summary statistics for wealth inequality (e.g., the median racial wealth gap)?

Chapter 3: Inequality Measurement — Setup

Input: An income frequency distribution $p(y)$ — for each income level $y \in [0, \bar{y}]$, $p(y)$ is the share of the population at that income level. This is the standard object in the inequality literature; the objects being ranked are *full distributions*, not individual outcomes.

Objective (Atkinson) measure. A social planner with welfare function $U(y)$ (increasing, concave) ranks income distributions by expected utility:

$$W^O = \int_0^{\bar{y}} U(y)\, p(y)\, dy.$$

The *equally distributed equivalent* income y_{EDE} satisfies $U(y_{EDE}) = W^O$. The inequality index is:

$$I^O = 1 - \frac{y_{EDE}}{\mu}, \qquad \mu = \text{mean income.}$$

Using the CRRA welfare function $U(y) = \frac{y^{1-\epsilon}}{1-\epsilon}$ (with $\epsilon \geq 0$ governing inequality aversion) yields a mean-independent, unit-free measure $I^O \in [0, 1]$.

Chapter 3: The Ambiguity-Based Measure

Reinterpreting wealth. A wealth distribution is modeled as a *state-contingent income distribution*: fix a state space $S = \{\text{Black}, \text{White}\}$ — the "birth lottery," in the spirit of Rawls' veil of ignorance — and let each act $f \in F$ assign to every state $s \in S$ an income frequency distribution, $f(s) = p(s)(y)$. The planner does not know the objective probability over states — this is *ambiguity* (Gilboa–Schmeidler 1989).

The ambiguity-averse welfare criterion takes the worst-case weighting over beliefs $\lambda \in \Delta(S)$:

$$W^A = \min_{\lambda \in \Delta(S)} \int_S \int_0^{\bar{y}} U(y)\, p(s)(y)\, dy\, d\lambda \;=\; \min_{s \in S} \mathbb{E}_{p(s)}\big[U(y)\big].$$

The minimum puts all weight on whichever racial group has the lower within-group expected utility that year.

The ambiguity EDE solves $U(w^A_{EDE}) = W^A$, giving the wealth inequality measure:

$$I^A = 1 - \frac{w^A_{EDE}}{\mu}.$$

Since $W^A \leq W^O$ by construction, $I^A \geq I^O$: the ambiguity criterion **systematically reports higher inequality** than the objective benchmark for the same wealth distribution.

Chapter 3: Data Pattern

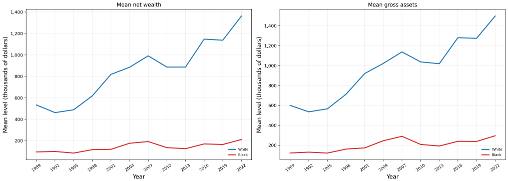
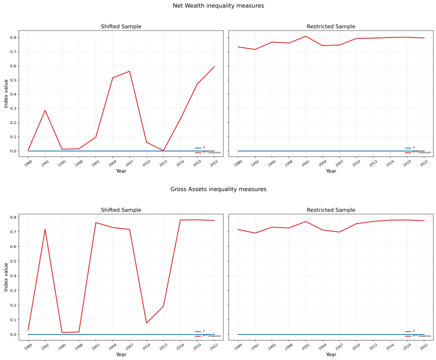


Figure 3.1: Mean net wealth (left panel) and mean gross assets (right panel) for Black and White households, SCF 1989–2022.

- Persistent, large racial wealth gap in both net wealth and gross assets.
- This is the empirical benchmark the inequality measures speak to.

Chapter 3: Ambiguity-Based Wealth Inequality



- The objective benchmark understates wealth inequality relative to the ambiguity-based measure for the same wealth distribution.
- This holds for both net wealth (top) and gross assets (bottom), and the measures track the broad time pattern of the median wealth-gap diagnostics.

Conclusion

- **Chapter 1**: return heterogeneity can match key wealth moments in a structural macro model, with an estimated distribution close to empirical evidence.
- **Chapter 2**: returns to net wealth exhibit a hump-shaped relationship with trust, and the maximizing level of trust lies in a moderate range between 6 and 7.
- **Chapter 3**: the ambiguity-based wealth inequality measure implies more inequality than the standard objective benchmark for the same wealth distribution.

Overall conclusion: heterogeneous returns are central both to explaining household wealth inequality and to evaluating it.